14. Explain the lipoproteins.

15. Give structure for lecithin and cephalin?

16. Give the classification of carbohydrates with examples.

17. Explain the functional diversity of proteins

18. Explain the biological functions of lipids.

19. Give an account on classification of lipids.

20. Explain the functions of DNA and RNA.

SUMMARY

Biomolecules are molecules that are present in all living organisms. It includes the macromolecules carbohydrates, proteins, lipids and nucleic acids. These are the four major classes of biomolecules

Carbohydrates

Carbohydrates are essential to maintain the life in both plants and animals. They are synthesized in plants by photosynthesis. Carbohydrates are good source of energy. Monosaccharides are simple sugars. They have a free aldehyde or keto group. Disaccharides are made of two monosaccharides connected by a glycosidic bond. Polysaccharides are the carbohydrates that yield more than ten monosaccharides units upon hydrolysis. They are further classified into homo polysaccharides and hetero polysaccharides based on the monomeric units. They are non-sugars and complex carbohydrates.

Proteins

Proteins are polymers of α-amino acids. These α-amino acids are joined together by the peptide bond which is formed in between the carboxyl group and amino group of successive amino acids. Proteins are formed from 20 different amino acids.

Proteins are classified into three types simple proteins, conjugated proteins and derived proteins. Simple protein yields only α-amino acids on hydrolysis. Conjugated protein yields α-amino acids and a non-protein part upon hydrolysis. The non-protein part is called the prosthetic group. Derived protein is formed by the action of acids, alkalis or enzymes on simple or conjugated proteins. These are partially hydrolyzed proteins.

Lipids

Lipids are generally esters of fatty acids and are building blocks of biological membranes. Lipid molecules hold a large amount of energy and are energy storage

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